Project Initialization and Planning Phase

Date	11 July 2026
Team ID	
Project Title	Rising waters - Flood Prediction using Machine Learning
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report
The proposed solution aims to predict flood occurrences using Machine Learning by analyzing historical rainfall and weather data. The system provides early flood risk predictions through a web application, helping disaster management authorities and citizens take preventive actions and reduce the impact of floods.

Project Overview	
Objective	The proposed solution aims to predict flood occurrences using Machine Learning by analyzing historical rainfall and weather data. The system provides early flood risk predictions through a web application, helping disaster management authorities and citizens take preventive actions and reduce the impact of floods.
Scope	The project processes weather and rainfall parameters, applies machine learning algorithms, and provides flood predictions through a simple web application accessible to users.
Problem Statement	
Description	Floods cause severe damage to lives, property, agriculture, and infrastructure. Existing warning systems often lack localized predictions and timely alerts.
Impact	An accurate flood prediction system enables early warnings, reduces disaster losses, improves emergency planning, and enhances public safety.
Proposed Solution	
Approach	Use historical rainfall and weather datasets to train machine learning models. Deploy the best-performing model through a Flask-based web application for real-time flood prediction.
Key Features	Machine Learning based flood prediction Weather and rainfall data analysis Real-time prediction through a web application Simple and user-friendly interface Early warning support for disaster preparedness

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU for model training and prediction	Intel Core i5 or above
Memory	RAM	8 GB
Storage	Project Files,datasets and Models	20 Gb Free Disk Space
Software		
Frameworks	Web frameworks	Flask
Libraries	Machine Learning & Data Processing	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Visual Studio code,Jupyter Notebook
Data		
Data	Historical rainfall and weather dataset	CSV dataset from Kaggle /Public Weather Dataset